

Amitkumar Pandey

THz Spectroscopy & Materials Science

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Physics researcher with a Master's degree from R.T.M. Nagpur University and hands-on research experience at the Tata Institute of Fundamental Research (TIFR) and the Variable Energy Cyclotron Centre (VECC). Specializes in Terahertz Time-Domain Spectroscopy (THz-TDS), Continuous Wave- Terahertz (CW-THz), spin dynamics and magnetism, and synthesis and characterization of polycrystalline materials. Co-author of seven research publications and manuscripts on terahertz spectroscopy of magnetic oxides and functional ceramics, with conference recognition including a Best Poster Award.

AREAS OF INTEREST

Terahertz Time-Domain Spectroscopy (THz-TDS) • Spin Dynamics & Magnetism • Material Synthesis (Polycrystalline Materials)

EDUCATION

2024 **M.Sc. Physics** — Department of Physics, R.T.M. Nagpur University, Nagpur (CGPA 7.60)

2022 **B.Sc. (Physics, Mathematics, Chemistry)** — Gondwana University, Gadchiroli (CGPA 7.86)

RESEARCH EXPERIENCE

Project Student / Assistant

Sept 2025 — Present

Tata Institute of Fundamental Research, Mumbai

Project Student

May — July 2023

Variable Energy Cyclotron Centre, Kolkata

RESEARCH PROJECTS

M.Sc. Dissertation — R.T.M. Nagpur University, Nagpur

July 2023 — April 2024

Calculation of the second virial coefficient of a Van der Waals gas and Hadron Resonance Gas. Supervisors: Dr. Jajati K. Nayak and Prof. U. A. Palikundwar.

Summer Project — Variable Energy Cyclotron Centre (VECC), Kolkata

May — July 2023

Understanding the virial coefficient in a Van der Waals gas and Hadron Resonance Gas. Supervisor: Dr. Jajati K. Nayak (VECC, Kolkata).

PUBLICATIONS

1. S. Tiwari, P. Singh, **A. Pandey**, A. K. Verma, S. Kumar, "Sustainable Synthesis of Biocompatible Superparamagnetic $\text{Zn}_{0.5}\text{Mn}_{0.5}\text{Fe}_2\text{O}_4$ Nanoparticles: Magnetic Field-Enhanced Antioxidant Response," *Journal of Magnetism and Magnetic Materials*, 2026, 174488. DOI: [10.1016/j.jmmm.2026.174488](https://doi.org/10.1016/j.jmmm.2026.174488) .
2. S. Joshi, **A. Pandey**, J. P. Parvathy, A. H. Kushwah, A. Mondal, S. S. Meena, S. Tiwari, S. S. Prabhu, "Low-Temperature Terahertz Spectroscopic Investigation on Hard Ferrimagnet CoFe_2O_4 ," *Journal of Infrared, Millimeter, and Terahertz Waves*, 47 (2026), 57. DOI: [10.1007/s10762-026-01163-y](https://doi.org/10.1007/s10762-026-01163-y) .
3. S. Tiwari, A. H. Kushwah, S. Joshi, **A. Pandey**, A. Mondal, A. Kshirsagar, S. Hazra, J. P. Parvathy, S. S. Prabhu, "Tuning Antiferromagnetic Resonance in NiO via Low-Level Zn Doping and Temperature," *Physica B: Condensed Matter*, 739 (2026), 418971. DOI: [10.1016/j.physb.2026.418971](https://doi.org/10.1016/j.physb.2026.418971) .
4. G. Pawar, **A. Pandey**, S. Joshi, J. P. Parvathy, N. S. Warade, P. B. Mane, S. S. Meena, S. Tiwari, S. S. Prabhu, "Structural, Magnetic and Terahertz Optical and Dielectric Properties of $\text{Zn}_{0.1}\text{Co}_{0.9}\text{Fe}_2\text{O}_4$," *Journal of Superconductivity and Novel Magnetism*, 39 (2026), 97. DOI: [10.1007/s10948-026-07194-9](https://doi.org/10.1007/s10948-026-07194-9) .

5. S. J. Helode, **A. Pandey**, S. J. Dhoble, “Structural, Thermogravimetric, and Optical Characterization of Er^{3+} Doped NaLaCaWO_6 Double Perovskite: Insights into Energy Transfer Mechanisms with Sm^{3+} and Eu^{3+} Ions,” *Ceramics International* (2025). DOI: [10.1016/j.ceramint.2025.09.349](https://doi.org/10.1016/j.ceramint.2025.09.349) .

Manuscripts Under Review

1. **A. Pandey**, S. Joshi, S. Tiwari, S. S. Prabhu, “Tuning Sub-THz Natural Ferromagnetic Resonances in CoFe_2O_4 by Cation Substitution,” *Ceramics International* (Under Review).
2. L. Darsi, S. Vaishnavi, S. Hazra, A. Mandal, **A. Pandey**, S. Ellipilli, G. Rana, S. S. Prabhu, “Experimental Demonstration of a High-Sensitivity THz Metamaterial Sensor for Breast Cancer Sensing,” *IEEE Sensors Journal* (Under Review).

CONFERENCES, WORKSHOPS & SCHOOLS

- **Aug 2026** — Poster Presentation, “Probing Magnetic Resonance in Ferrimagnets and Antiferromagnets via Terahertz Time-Domain Spectroscopy,” National Symposium on Intense Laser Matter Interaction 2026, IIT (ISM) Dhanbad — **Best Poster Award**.
- **Feb 2025** — Poster Presentation, “Spectroscopic Investigations of Novel Sm^{3+} Activated NaLaCaWO_6 Phosphors for Lighting Applications,” National Conference on Innovation in Science and Technology for Sustainable Development, Nagpur.
- **July 2024** — Summer School on Gravitational-Wave Astronomy, ICTS, Bangalore.
- **June 2024** — STEM Tinkering Workshop, Science Activity Centre — IISER, Pune.
- **Sept 2023** — Aditya-L1 Fifth Support Cell Workshop, IIT Kanpur.
- **June 2022** — Summer Course on Curves and Surfaces, ICTS, Bangalore.
- **May 2022** — Introductory Summer School on Astronomy & Astrophysics, IUCAA, Pune.

SKILLS

Instrumentation

- Photoluminescence Spectroscopy (PL/PLE)
- THz Time-Domain Spectroscopy — temperature- and polarization-dependent
- Continuous-wave THz Spectroscopy — temperature-dependent

Computational

LaTeX, Origin, Rietveld Refinement

REFEREES

Prof. Shriganesh Prabhu

Dept. of Condensed Matter Physics & Material Science, TIFR, Mumbai

Prof. U. A. palikundwar

Department of Physics, R.T.M. Nagpur University

Dr. Jajati K. Nayak

Scientific Officer (G), Theoretical Nuclear & Many Body Physics Division, VECC, Kolkata